



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

10 / 10 completed

Integer Array

INTEGER ARRAY

"Write a C program to demonstrate pointer arithmetic on an integer array. Display element values using both array indexing and pointer notation. Show address difference between consecutive elements.

Sample Input:

Enter number of elements: 4

Enter elements:

10 20 30 40

Sample Output:

Using Array Indexing:

arr[0] = 10

arr[1] = 20

arr[2] = 30

arr[3] = 40

Using Pointer Notation:

*(ptr+0) = 10

*(ptr+1) = 20

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 void main(){
3     int n,i;
4     printf("enter the no of elements:\n");
5     scanf("%d",&n);
6     int arr[n];
7     int*ptr=arr;
8     printf("enter %d elements:\n");
9     for(i=0;i<n;i++){
10         scanf("%d",&arr[i]);
11     }
12     printf("\n array indexing:\n");
13     for(i=0;i<n;i++){
14         printf("arr[%d]= %d\n",i,arr[i]);
15     }
16 }
17 printf("\n using pointer notation\n");
18 for(i=0;i<n;i++){
19     printf("*(ptr+d)= %d\n",i,*(ptr+i));
20 }
21
```

OUTPUT

enter the no of elements:

enter 4 elements:

array indexing:

arr[0]= 10

arr[1]= 20

arr[2]= 30

arr[3]= 40

using pointer notation

INPUT

4

10 20 30 40



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